Student: Xirong Zhang **Date:** 11/30/25	**Instructor:** Dmitriy Ivanov **Course:** Math 120 Fall 2025	**Assignment:** Section 1.4

1. Compute the product using (a) the definition where A**x** is the linear combination of the columns of A using the corresponding entries in **x** as weights, and (b) the row-vector rule for computing A**x**. If a product is undefined, explain why.

$$\begin{bmatrix} -8 & 6 \\ 9 & 1 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} 10 \\ -1 \\ 10 \end{bmatrix}$$

(a) Compute the product using the definition where A**x** is the linear combination of the columns of A using the corresponding entries in **x** as weights. If the product is undefined, explain why. Select the correct choice below and, if necessary, fill in any answer boxes to complete your choice.

- **A.** A**x** =
- **B.** The matrix-vector A**x** is not defined because the number of rows in matrix A does not match the number of entries in the vector **x**.
- **C.** The matrix-vector A**x** is not defined because the number of columns in matrix A does not match the number of entries in the vector **x**.

(b) Compute the product using the row-vector rule for computing A**x**. If the product is undefined, explain why. Select the correct choice below and, if necessary, fill in any answer boxes to complete your choice.

- **A.** A**x** =
- **B.** The matrix-vector A**x** is not defined because the row-vector rule states that the number of rows in matrix A must match the number of entries in the vector **x**.
- **C.** The matrix-vector A**x** is not defined because the row-vector rule states that the number of columns in matrix A must match the number of entries in the vector **x**.

2. Compute the product using the methods below. If a product is undefined, explain why.
a. The definition where A**x** is the linear combination of the columns of A using the corresponding entries in **x** as weights.
b. The row-vector rule for computing A**x**.

$$\begin{bmatrix} 6 & 5 \\ -4 & -3 \\ 7 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -4 \end{bmatrix}$$

a. Set up the linear combination of the columns of A using the corresponding entries in **x** as weights. Select the correct choice below and, if necessary, fill in any answer boxes to complete your choice.

A.

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = (x_1)\mathbf{a}_1 + (x_2)\mathbf{a}_2 = (\boxed{2})\begin{bmatrix} 6 \\ -4 \\ 7 \end{bmatrix} + (\boxed{-4})\begin{bmatrix} 5 \\ -3 \\ 4 \end{bmatrix}$$

(Type the terms of your expression in the same order as they appear in the original expression.)

B. $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = (x_1)\mathbf{a}_1 + (x_2)\mathbf{a}_2 + (x_3)\mathbf{a}_3 = (\quad)\quad + (\quad)\quad + (\quad$
(Type the terms of your expression in the same order as they appear in the original expression.)

C. The matrix-vector A**x** is not defined because the number of columns in matrix A does not match the number of entries in the vector **x**.

D. The matrix-vector A**x** is not defined because the number of rows in matrix A does not match the number of entries in the vector **x**.

b. Set up the product A**x** using the row-vector rule. Select the correct choice below and, if necessary, fill in the answer boxes to complete your choice.

A.

$$\begin{bmatrix} \boxed{2 \cdot 6 - 4 \cdot 5} \\ \boxed{-4 \cdot 2 - 3 \cdot (-4)} \\ \boxed{2 \cdot 7 - 4 \cdot 4} \end{bmatrix}$$ (Do not simplify.)

B.

$$\begin{bmatrix} \boxed{} \\ \boxed{} \end{bmatrix}$$ (Do not simplify.)

C. The matrix-vector A**x** is not defined because the row-vector rule states that the number of columns in matrix A must match the number of entries in the vector **x**.

D. The matrix-vector A**x** is not defined because the row-vector rule states that the number of rows in matrix A must match the number of entries in the vector **x**.

Evaluate the expressions found in the previous steps. Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

A.

$$A\mathbf{x} = \begin{bmatrix} -8 \\ 4 \\ -2 \end{bmatrix}$$ (Simplify your answer.)

B. The matrix-vector A**x** is not defined.

3. Use the definition of A**x** to write the vector equation as a matrix equation.

$$x_1\begin{bmatrix}-4\\-1\\5\\-8\end{bmatrix}+x_2\begin{bmatrix}-8\\-1\\5\\-8\end{bmatrix}+x_3\begin{bmatrix}7\\-2\\-9\\-2\end{bmatrix}=\begin{bmatrix}5\\3\\7\\1\end{bmatrix}$$

The matrix equation is $\begin{bmatrix}-4&-8&7\\-1&-1&-2\\5&5&-9\\-8&-8&-2\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}5\\3\\7\\1\end{bmatrix}$

4. Use the definition of A**x** to write the vector equation as a matrix equation.

$$z_1\begin{bmatrix}-5\\5\end{bmatrix}+z_2\begin{bmatrix}3\\-4\end{bmatrix}+z_3\begin{bmatrix}0\\4\end{bmatrix}+z_4\begin{bmatrix}-4\\-1\end{bmatrix}=\begin{bmatrix}0\\6\end{bmatrix}$$

The vector equation written as a matrix equation is $\begin{bmatrix}-5&3&0&-4\\5&-4&4&-1\end{bmatrix}\begin{bmatrix}z_1\\z_2\\z_3\\z_4\end{bmatrix}=\begin{bmatrix}0\\6\end{bmatrix}$.

5. Given A and **b** to the right, write the augmented matrix for the linear system that corresponds to the matrix equation A**x** = **b**. Then solve the system and write the solution as a vector.

$$A=\begin{bmatrix}1&3&-4\\-2&-2&0\\-4&-7&2\end{bmatrix},\ \mathbf{b}=\begin{bmatrix}-8\\-4\\-13\end{bmatrix}$$

Write the augmented matrix for the linear system that corresponds to the matrix equation A**x** = **b**.

$$\begin{bmatrix}1&3&-4&-8\\-2&-2&0&-4\\-4&-7&2&-13\end{bmatrix}$$

Solve the system and write the solution as a vector.

$$\mathbf{x}=\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}-3\\5\\5\end{bmatrix}$$

6. Let $\mathbf{u}=\begin{bmatrix}-25\\16\\7\end{bmatrix}$ and $A=\begin{bmatrix}5&-7\\-2&4\\1&1\end{bmatrix}$. Is **u** in the plane in $\mathbb{R}^3$ spanned by the columns of A? Why or why not?

Select the correct choice below and fill in the answer box to complete your choice.
(Type an exact answer for each matrix element.)

A. Yes, multiplying A by the vector $\begin{bmatrix}2\\5\end{bmatrix}$ writes **u** as a linear combination of the columns of A.

B. No, the reduced echelon form of the augmented matrix is _____, which is an inconsistent system.

7. Let $\mathbf{u} = \begin{bmatrix} -4 \\ -2 \\ 7 \end{bmatrix}$ and $A = \begin{bmatrix} 5 & 8 & 9 \\ 1 & 2 & 1 \\ 3 & 1 & 13 \end{bmatrix}$. Is **u** in the subset of $\mathbb{R}^3$ spanned by the columns of A? Why or why not?

Select the correct choice below and fill in the answer box to complete your choice.
(Type an integer or decimal for each matrix element.)

A. No, the reduced echelon form of the augmented matrix is $\begin{bmatrix} 1 & 0 & 5 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, which is an inconsistent system.

B. Yes, multiplying A by the vector _____ writes **u** as a linear combination of the columns of A.

8. Let $A = \begin{bmatrix} 1 & -4 & -3 \\ -4 & 4 & 0 \\ 2 & 4 & 6 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Show that the equation $A\mathbf{x} = \mathbf{b}$ does not have a solution for all possible **b**, and describe the set of all **b** for which $A\mathbf{x} = \mathbf{b}$ does have a solution.

How can it be shown that the equation $A\mathbf{x} = \mathbf{b}$ does not have a solution for all possible **b**? Choose the correct answer below.

A. Find a vector **x** for which $A\mathbf{x} = \mathbf{b}$ is the zero vector.

B. Row reduce the matrix A to demonstrate that A has a pivot position in every row.

C. Row reduce the matrix A to demonstrate that A does not have a pivot position in every row.

D. Find a vector **b** for which the solution to $A\mathbf{x} = \mathbf{b}$ is the zero vector.

E. Row reduce the augmented matrix $\begin{bmatrix} A & \mathbf{b} \end{bmatrix}$ to demonstrate that $\begin{bmatrix} A & \mathbf{b} \end{bmatrix}$ has a pivot position in every row.

Describe the set of all **b** for which $A\mathbf{x} = \mathbf{b}$ does have a solution.

$0 = b_2 + 2b_1 + b_3$

(Type an expression using b_1, b_2, and b_3 as the variables and 1 as the coefficient of b_3.)

9. Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}$. Does $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ span $\mathbb{R}^4$? Why or why not?

Choose the correct answer below.

A. Yes. When the given vectors are written as the columns of a matrix A, A has a pivot position in every row.

B. No. When the given vectors are written as the columns of a matrix A, A has a pivot position in only three rows.

C. Yes. Any vector in $\mathbb{R}^4$ except the zero vector can be written as a linear combination of these three vectors.

D. No. The set of given vectors spans a plane in $\mathbb{R}^4$. Any of the three vectors can be written as a linear combination of the other two.

10. Determine whether the statement below is true or false. Justify the answer.

A vector **b** is a linear combination of the columns of a matrix A if and only if the equation A**x** = **b** has at least one solution.

Choose the correct answer below.

- **A.** This statement is false. If the equation A**x** = **b** has infinitely many solutions, then the vector **b** cannot be a linear combination of the columns of A.
- **B.** This statement is true. The equation A**x** = **b** has the same solution set as the equation $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$.
- **C.** This statement is false. If the matrix A is the identity matrix, then the equation A**x** = **b** has at least one solution, but **b** is not a linear combination of the columns of A.
- **D.** This statement is true. The equation A**x** = **b** is unrelated to whether the vector **b** is a linear combination of the columns of a matrix A.